

CSE 310 Online (C1): Python Dictionary

Time: 40 minutes Full Marks: 10

You will create a new .l file, which must not be reused from your offline lexical-analysis assignment, to validate a simplified Python dictionary. The input consists of a single dictionary containing key–value pairs.

A valid dictionary follows these rules:

- The complete dictionary is enclosed within { and }.
- A key may be a quoted string or an integer constant.
- A quoted string may use either single quotation marks or double quotation marks.
- Each key is followed by a colon :
- A value may be a quoted string, an integer constant, True, False, None, or another dictionary.
- A comma , separates consecutive key–value pairs.
- A comma after the final key–value pair is optional.

For example, the following input is valid:

```
{
    'name': "alice",
    101: {
        "city": 'sylhet',
        "active": True
    },
    "phone": None
}
```

Your task is to validate:

- proper matching and nesting of { and };
- proper opening and closing of single and double quotation marks;
- valid string or integer keys;
- correct use of : between a key and its value; and
- correct use of , between consecutive key–value pairs.

You may use Flex states to track whether the scanner is reading a key, a value, or a quoted string. A counter or a built-in C/C++ data structure may be used to track nested dictionaries.

Required error checking

Report only the first error found. Use the following messages where applicable.

- If a key is neither a quoted string nor an integer, report:

Error: invalid dictionary key

- If two key–value pairs are not separated by ,, report:

Error: missing pair separator

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- If an opening brace remains unmatched at the end of input, report:

Error: unmatched {

Sample I/O

Sample 1: Valid Python dictionary

Input

```
{
    'name': "alice",
    101: { "city": 'sylhet', "active": True },
    "phone": None
}
```

Output

Valid Python Dictionary

Sample 2: Invalid unquoted key

Input

```
{ name: "alice", "city": "dhaka" }
```

Output

Error: invalid dictionary key

Sample 3: Missing pair separator

Input

```
{ "name": "alice" "city": "dhaka" }
```

Output

Error: missing pair separator

Sample 4: Unmatched opening brace

Input

```
{ "name": "alice", "address": { "city": "sylhet" }
```

Output

```
Error: unmatched {
```

Marks Distribution

- Addressing valid dictionaries: **6 marks**
- Error checking: **4 marks**

Submission: Submit only your 2205ABC.l file. Do not submit lex.yy.c, executables, or output files.